Type . . .

Truck | Dinosaur

Entropy $= -\frac{1}{2}\log_2(1) + 0 = 0$

Truck: $p(yes) = 1$, $p(no) = 0$

Dinosaur: ~~$p(no) = 0$~~ $p(yes) = 0$, $p(no) = 1$

Entropy = 0

Information Gain = 0.971

③ Hence, the maximum information gain comes from splitting the type of toys.

for size .

Entropy :

Big | Small

Big: yes : 1, no = 0, $e = 0$

Small: yes : $\frac{1}{3}$, no : $\frac{2}{3}$

$e = -\frac{1}{3}\log_2\left(\frac{1}{3}\right) - \frac{2}{3}\log_2\left(\frac{2}{3}\right)$

0.918

Information gain $= 0.971 - 0 - \frac{3}{5}\,0.918 = 0.020$ bits

(1) $p(\cancel{\text{Is tur}}\ \text{yes}) = \frac{3}{5}$

$p(\text{no}) = \frac{2}{5}$

Entropy $= -\frac{3}{5}\log_2\left(\frac{3}{5}\right) - \frac{2}{5}\log_2\left(\frac{2}{5}\right)$

$= 0.971$ bits

(2) $p(\text{Red}) = \frac{3}{5}$ Blue $= \frac{2}{5}$

~~Information gain~~ :

~~size p(big)~~ p(yes) =

Red

Entropy = 0.918

yes $= \frac{2}{3}$

no $= \frac{1}{3}$

Blue

Entropy = 1

yes $= \frac{1}{2}$

no $= \frac{1}{2}$

Information gain $= 0.971 - \frac{3}{5}0.918 - \frac{2}{5}1 = 0.020$ bits